

## **Proof of genuine engagement: blockchain and the neo-slacktivism problem**

Final Decision: **Accept with Minor Revision (1&2)+ Reorganization of manuscript (3)**

### Reviewer level of knowledge on the topic:

My own evaluation is

- High on Blockchains & Tokenomics
- Medium on the specific area of Blockchains based Social-Media
- Almost general knowledge on Slacktivism, Structural and Institutional theories.

### My Detailed Review:

#### **1-There is some scientific unclarity in the figures:**

- Figure 4 accurate explanation appears only in section 10, the text before it is kind of vague, and Fig.7 seems a natural follow of it as the big picture of it.
- Fig.5 does not demonstrate the error described with just one sentence in the accompanying text. Also, there is no explanation what colors mean? It should be written either within the graph or in the text: why do scenarios 1&3 have dotted frame? Why scenario1 is in gray? why some text is in red; is P1 here detected as fake?
- In Fig.6 what I get from the text is that PTs are possible states or (support/oppose/...) toward a PL; hence, they're not a ledger themselves to be connected as a chain in the figure.

**2- The last paragraph of section 10 should be transformed into a detailed subsection discussing failure cases (2018-2019); the reasons and how can be mitigated or avoided** in light of newer following ones that succeeded up till now. Some references are embedded in the manuscript; also, papers criticizing/analyzing Community on X since the authors used it as a basis of Comparison. (The authors do not have to cite all embedded references; just go through all of them to develop a global understanding, then cite what is relevant to the discussion)

-That's my recommendation (the blockchain dimension and analysis); the editors or the authors may add to it or prefer the political science approach alone (I'm not an expert on that, see the "PS." After my review).

#### **3-The paper is still not well-organized enough:**

- This comment from the previous report "*In some parts, the paper lacks organization; sometimes there 2 or more pages of paragraphs that seem like a long essay*" is not yet completely fixed.
- You read some lines and get the feeling "I've read that before, is this repeated?", you scroll back up and sometimes find it's lightly mentioned before it is detailed (which could be acceptable), but sometimes you find it is like a repeated cliché in many parts.
- The paper does not contain a roadmap of all sections in the first introductory section as most papers do.
- If tried to do it as in online publications sites, you will find the following:

**(I needed to do that after the first read, because I felt lost in a pretty long essay)**

- 1) Introduction
  - a) Conceptual definitions
  - b) AI driven neo-slacktivism
- 2) Blockchain and its Affordance
- 3) Understanding the methodology
  - a) interview meets
  - b) focus groups
  - c) analysis of data and sampling approaches
  - d) Coding of data **(I'm not sure this is the accurate title, you are not "coding" you're deducing info from data or comprehending; if you are using the terminology of "*coding schema*", check the PS. After the final decision)**
- 4) Theoretical Approach
- 5) Blockchains and Tracking activities **(I suggested subsections in the text ~ 3 pages of continuous text)**
- 6) Blockchain and Data Management
- 7) Potential Benefits of Blockchains
- 8) Algorithmic Blockchain Driven Data Management .... **(too long title here & in many places + check important notes on Fig.4&5)**
- 9) Projecting Key Risks
  - a) Regulatory oversight
  - b) Systematic barriers

## 10) Discussion

- a) Cross-referenced ledgers (this finally explains accurately Fig.4, why it is delayed till here?! maybe moved 6)Architecture)
- b) Tiered consensus schemes
- c) Token economics
- d) Decentralized moderation (even Fig.7 is the large picture of Fig.4 which is kind of zooming into a specific detail)
- e) Last paragraph needs to be separated as “Risks and/or Challenges” with some mitigations, could also precede or follow it with “10.5 Limitations” for the part I suggested to be entitled “5.1 Blockchain Role & Limitations” on how blockchains can’t fix everything; in the whole picture maybe here seems better. I mean in the final discussion with the failure examples is a better place to say what means do not blame the blockchain if you chose wrong incentives for the significant actions, or wrong AI rules to define what is genuine, ..... (ex. <https://www.stern.nyu.edu/experience-stern/faculty-research/who-moderates-social-media-giants-call-end-outsourcing>)

## 11) Conclusion

### Final decision:

I do not believe in rejecting a paper just for text organization or writing problems if scientifically good enough; I’m willing for a third-round review, if the authors resolved the remaining issues.

### PS.:

I assume the journal had another reviewer to investigate the political science part. However, when I double checked with Gemini and Copilot to check if I’ve missed any error, they commented on them as follows: (their reports are attached at the tail of the report)

- Copilot said on the methodology: “*The thematic analysis is described but not demonstrated (no coding schema, no theme development process)*”. For me on the technical side, the authors simply used the zoom meetings to develop an understanding that helped them propose the architectures and write the paper; although, I did not see a list of questions with how many agreed and/or disagreed as other papers that depend on asking a sample of humans (ex.:
- Copilot also: “*The paper needs clearer positioning within democratic theory and civic participation literature*”. I believe that’s for the editors to decide if that suits JeDEM style, because to me the paper is more technical than political. The rest are about details already written but missed by Copilot because the writing is not well organized.
- Gemini recommends *applying the institutional theory constraints e.g. normative pressures when discussing the failure of previous protocols* meaning last paragraph of 10.4; for me I recommend adding suggested mitigations whether your own, from literature, or example networks)

## Copilot Review Report:

Below is a **corrected, fair, and balanced JeDEM-style review report**, written from the perspective of a reviewer who has read the full manuscript.

### **Corrected JeDEM Review Report (Reviewer Perspective)**

*(Fair, accurate, and acknowledging interview quotes + limitations section)*

#### **Manuscript Title**

*Proof of genuine engagement: blockchain and the neo-slacktivism problem*

#### **Overall Recommendation**

#### **Major Revision**

The manuscript is timely, ambitious, and highly relevant to JeDEM's focus on digital governance, civic participation, and socio-technical systems. It addresses the emerging problem of AI-driven neo-slacktivism and proposes blockchain-based civic ledgers, metadata identifiers, and Proof-of-Humanity systems as mechanisms to authenticate human engagement.

The paper is clearly agenda-setting and offers several original conceptual contributions. It also includes interview quotations and a thoughtful limitations section, which significantly strengthen the empirical and critical dimensions. However, substantial revisions are needed to sharpen the theoretical framing, integrate empirical findings more tightly, and streamline the structure.

### **1. Contribution and Relevance**

#### **Strengths**

- The manuscript tackles a novel and urgent issue: *"AI-driven neo-slacktivism... amplified and manipulated by machine learning algorithms"*.
- The conceptual model linking blockchain affordances to civic engagement is innovative and well-illustrated.
- Interview quotations provide authentic practitioner perspectives and demonstrate genuine empirical engagement.
- The limitations section is strong and explicitly acknowledges that *blockchain cannot solve everything*, which is essential for JeDEM's standards.
- Section 10 offers a sophisticated discussion of token-value volatility, governance deadlocks, incentive misalignment, and decision-making delays — demonstrating a mature understanding of blockchain's socio-technical constraints.

#### **Weaknesses**

- The conceptual contribution is promising but not yet sharply articulated.
- The paper needs clearer positioning within democratic theory and civic participation literature.
- The empirical findings, while present, are not fully integrated into the conceptual model.

**Verdict:** Strong potential, but the contribution must be sharpened and more tightly linked to democratic governance theory.

### **2. Structure and Clarity**

#### **Strengths**

- The manuscript is well-organized and includes clear sections on interviews, focus groups, and conceptual diagrams.
- The figures (e.g., metadata-driven civic ledger) are helpful and visually strong.
- The limitations section and Section 10 add important balance.

#### **Weaknesses**

- The introduction is overly long and contains extensive background material that could be condensed.
- Some sections read more like literature reviews than analytical development.
- The conceptual model is described narratively but not formalized.

**Required Action:** Condense early sections and foreground the blockchain–neo-slacktivism argument more quickly.

### **3. Literature Review**

#### **Strengths**

- The review is impressively broad and up-to-date.
- It covers slacktivism, metadata ecosystems, synthetic data, blockchain governance, DAOs, and Proof-of-Humanity.

#### **Weaknesses**

- The review is too expansive and occasionally unfocused.

- Key JeDEM-relevant literature on digital democracy, civic tech, and participatory governance is underrepresented.
- The political science dimension needs strengthening.

**Required Action:** Reduce technical digressions and integrate more democratic theory.

#### 4. Methodology

##### Strengths

- The methodology section is detailed and includes:
  - purposive sampling,
  - two waves of interviews,
  - two focus groups,
  - thematic analysis,
  - stakeholder comparison,
  - conceptual network mapping.
- Interview quotations significantly strengthen the empirical foundation.

##### Weaknesses

- The thematic analysis is described but not demonstrated (no coding schema, no theme development process).
- Interview quotes should be more explicitly tied to conceptual claims. **When section 5 is divided to subsections, they will be linked to the correct text.**
- The link between empirical insights and the proposed blockchain model needs clearer articulation.

**Required Action:** Show how interview data *produced* the conceptual insights, not just how it *supports* them.

#### 5. Findings and Discussion

##### Strengths

- The paper proposes a compelling idea: blockchain-based civic ledgers could make “*inaction visible, rewarding depth of interaction over breadth*”.
- Interview quotations enrich the findings and demonstrate practitioner perspectives.
- Section 10 provides a sophisticated discussion of:
  - token-value volatility,
  - governance deadlocks,
  - incentive misalignment,
  - decision-making delays,
  - risks of token-based participation metrics.

These are excellent additions that elevate the manuscript beyond techno-optimism.

##### Weaknesses

- The findings are still more conceptual than empirical.
- The discussion could better integrate interview insights with the proposed model. **I agree the authors should have added interviewers’ comments about the failure cases if existed, but this part was added only after the first review. The authors can compensate that by checking the available material whether in literature or credible crypto blogs and/or discussion boards.**
- The paper should more explicitly address: **I’ve already recommended subsection 10.5 or 10.6 backed by examples from the failure cases; currently this is discussed in the first paragraph of section 5 and last paragraph of section 10**
  - privacy risks,
  - exclusion risks,
  - regulatory constraints,
  - potential misuse of blockchain in activist contexts.

**Required Action:** Expand the critical discussion and integrate empirical evidence more tightly.

#### 6. Limitations Section

##### Strengths

- The manuscript acknowledges that blockchain is not a panacea.
- The limitations section correctly notes:

- blockchain cannot solve all trust problems,
- token-based incentives can distort behavior,
- governance may become deadlocked,
- token value fluctuations affect participation,
- decision-making may slow down or stall.

These are essential for JeDEM's standards of critical reflection.

**Weaknesses** kind of repeating the previous weakness

- The limitations could be more systematically structured.
- Some limitations are scattered across sections rather than consolidated.

**Required Action:** Consolidate all limitations into a single coherent section.

## 7. Writing Quality

**Strengths**

- The writing is polished, articulate, and stylistically sophisticated.
- The manuscript demonstrates strong domain knowledge.

**Weaknesses**

- Some sections are overly dense.
- Occasional rhetorical flourishes reduce academic precision.

## 8. Specific Comments

**Major Issues**

1. Clarify the theoretical contribution and formalize the conceptual model.
2. Strengthen the integration of interview quotations into the argument.
3. Condense background sections.
4. Expand the critical discussion of risks and governance constraints.
5. Consolidate limitations into a single section.

**Minor Issues**

- Define “token exhaustion” more precisely.
- Improve transitions between sections.
- Ensure consistent terminology.

**Final Recommendation: Major Revision**

The manuscript is promising, original, and highly relevant to JeDEM. The presence of interview quotations and a thoughtful limitations section significantly strengthens the work. With revisions to sharpen the theoretical contribution, integrate empirical findings, and streamline the structure, the paper could become a strong agenda-setting contribution to digital governance scholarship.

# Gemini Review Report:

## Peer Review Report

**Journal:** [eJournal of eDemocracy and Open Government \(JeDEM\)](#)

**Manuscript Title:** Proof of genuine engagement: blockchain and the neo-slacktivism problem

**Paper Category:** Research Paper

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### 1. Overall Evaluation and Recommendation

The manuscript addresses a highly contemporary problem: **AI-driven neo-slacktivism** and the erosion of authentic human participation in digital public spaces. It explores how blockchain and decentralized ledger technologies (DLT) can verify human civic actions. The topic aligns perfectly with *JeDEM's* scope.

**Recommendation: Major Revision.** The paper possesses a robust conceptual baseline and empirical depth but suffers from structural density, self-citation saturation, and technical idealized assumptions that need tempering.

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### 2. Relevance to eDemocracy and Open Government

The manuscript is **highly relevant** to the themes of *JeDEM*. It moves beyond traditional clicktivism to examine how large language models (LLMs) pollute public metrics. The proposed architectural systems directly influence:

- **e-Participation and e-Voting:** Verifying unique human intent in digital processes.
  - **Data Governance and Transparency:** Shifting from centralized proprietary platform control to verifiable public ledgers.
  - **Civic Trust:** Mitigating synthetic content manipulation and bot ecosystems.
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### 3. Theoretical Framework

The authors ground their analysis in **Institutional Theory**, highlighting coercive, normative, and mimetic pressures.

- **Strength:** Effectively demonstrates how profit-driven algorithms conflict with decentralized accountability.
  - **Weakness/Revision Needed:** The transition from Institutional Theory to the specific socio-technical mechanics of blockchain requires clearer mapping. Section 4 is brief and reads isolated from the technical architectures described in Sections 7 and 8. The author should weave institutional inertia concepts directly into the discussion of the "triple-layer verification" model.
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### 4. Methodology

The empirical core consists of **31 semi-structured interviews** across two waves and **two focus group discussions** featuring 26 distinct participant entries.

- **Commendation:** Purposive sampling targeting GitHub contributors and excluding speculative traders is an excellent choice to isolate governance value.
  - **Critical Flaw:** The text introduces multiple architectural diagrams (Figures 4, 5, 6, 7) labeled as "Source: own illustration based on network analysis of focus groups". However, the methodology does not explain *how* a qualitative focus group discussion translated into exact programmatic block designs (e.g., specific parameters like  $\backslash(VR_{CL})$  or  $\backslash(PLx.y^{\{MT\}})$ ). The authors must explain whether these models were proposed *by* the participants or developed independently to interpret qualitative results. **For me I disagree with that**
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### 5. Presentation and Visualizations **kind of disagree and added my comments on the figures**

The paper includes several dense technical diagrams:

- **Figures 1 & 2:** Conceptually clear and establish standard blockchain transaction trails.
  - **Figures 4, 5, 6, & 7:** Visually anchor the conceptual framework but are excessively repetitive. Figure 7 duplicates elements of Figure 2 and Figure 6.
  - **Revision Requirement:** Synthesize Figures 4, 5, and 7 into a single, cohesive framework layout. This will free up valuable text real estate and improve reading flow.
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### 6. Validity and Critical Discussion

The paper hits a wall of **technological idealism** that requires grounding:

- **Token Exhaustion vs. Accessibility:** The authors argue that consuming non-transferable civic tokens stops bots through resource depletion. They fail to analyze how this prevents low-income or marginalized humans from sustaining participation, contradicting the stated benefit of lowering entry barriers.
  - **The Reality Paradox:** Section 10.4 admits that historical protocols attempting content verification (e.g., *Dirt* and *Pressland*) collapsed in the early 2020s, and that centralized corporate mechanisms (e.g., *Community Notes on X*) remain empirically superior. This major admission needs to be front-and-center, rather than buried at the very end. The discussion must directly evaluate *why* decentralized solutions fail in implementation compared to corporate ones.
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## 7. References and Citation Balance

A major structural concern is the citation pattern: **Disagree, we are the one asked for the new 2026 refs. + 3 out of 103 ref.s is less than 3%+ I don't think Kassen is a co-author due to the organizational mess of the manuscript**

- **Self-Citation Density:** Citations to a single author profile (Kassen 2025, 2026a, 2026b) appear continuously across multiple foundational arguments.
  - **Revision Requirement:** The literature review must balance this out by integrating broader, independent scholars exploring decentralized identities, decentralized autonomous organizations (DAOs), and digital sociology to minimize bias.
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## 8. Concrete Revisions Required for Acceptance

1. **Methodological Clarification:** Explicitly detail the link between qualitative transcript codes and the technical design choices in the architectural models.
  2. **Consolidate Visuals:** Merge overlapping figures to improve presentation and remove visual redundancy.
  3. **Address Equity Concerns:** Add a dedicated subsection discussing whether civic token expenditure risks disenfranchising resource-poor human participants.
  4. **De-silo Theory:** Explicitly apply the institutional theory constraints (e.g., normative pressures) when discussing the structural failure of past protocols like *Dirt*. **Here is where I have different comments, but out of my expertise if this is valid too.**
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